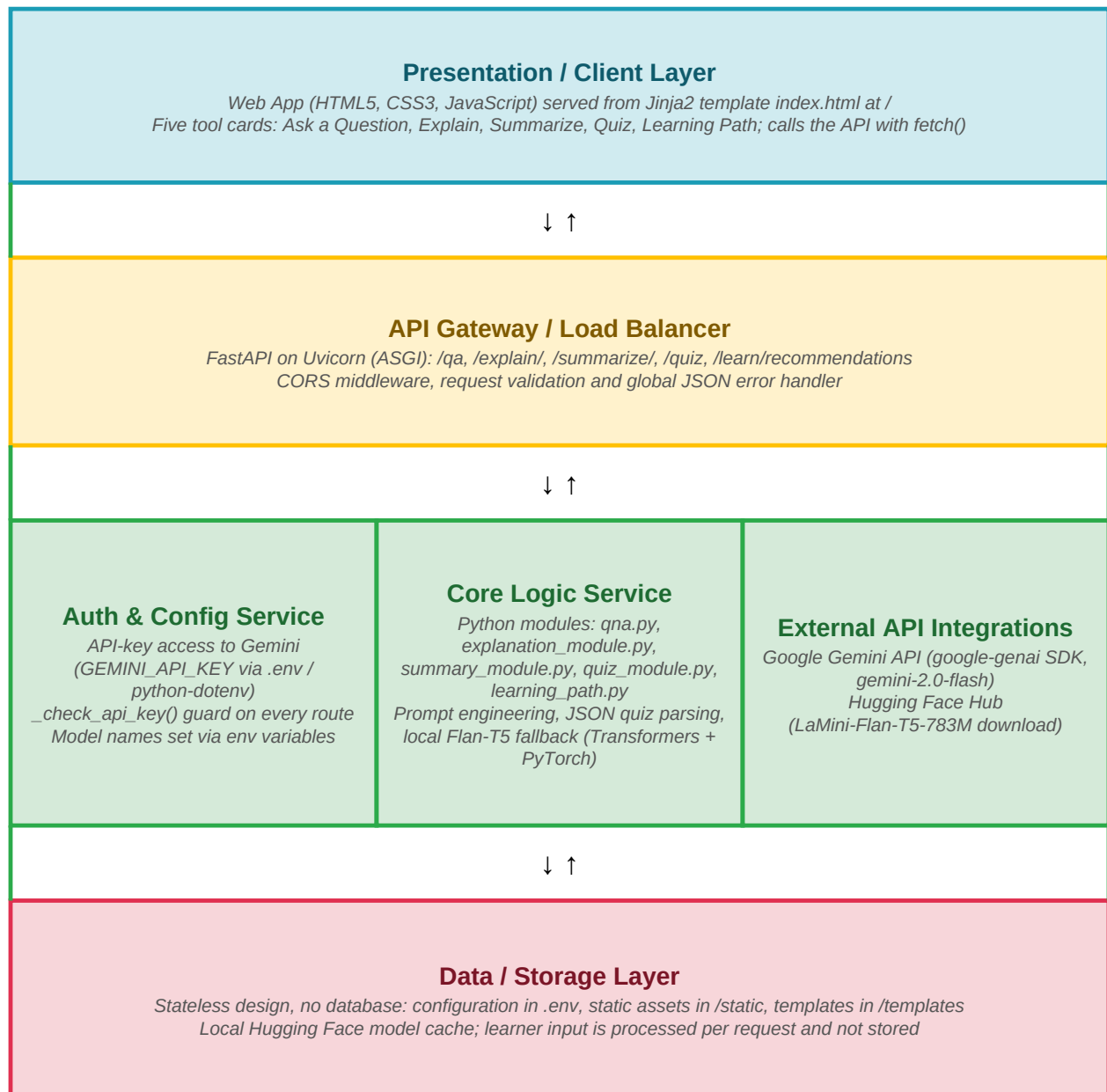


Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie - AI Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Single-page web interface where learners ask questions, request explanations, paste text to summarise, generate quizzes and get learning paths. Sends requests to the API and renders answers and interactive quiz options.	HTML5, CSS3, JavaScript (fetch API), Jinja2 template
API Gateway	Single entry point that exposes the REST endpoints, validates input, applies CORS rules, routes each request to the right module and returns JSON errors.	FastAPI, Uvicorn (ASGI), CORS middleware
Auth & Config Service	Loads settings from environment variables and checks that a valid Gemini API key is configured before any AI request runs.	python-dotenv, .env, API-key check
Microservices (Core Logic Modules)	Independent modules for Q&A, explanation, summarisation, quiz generation and learning paths. Each builds a task-specific prompt, calls the model and cleans the output (e.g. parses quiz JSON). Explanations can run on a local model when it is available.	Python 3.13, google-genai, Transformers, PyTorch, sentencepiece
External API Integrations	Cloud AI that generates the answers, plus the model hub used to fetch the local fallback model.	Google Gemini API (gemini-2.0-flash), Hugging Face Hub (LaMini-Flan-T5-783M)
Database	The app is stateless and has no database: configuration, static files and templates are kept on disk, the local model is cached, and learner input is not stored.	.env, /static, /templates, Hugging Face local cache